

CSMFAB000000000113-Vol-16

Trajectory Engineering: Optimal Launch Arcs and Suborbital Path Optimization

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1. Two-Point Boundary Value Problem (TPBVP)

The optimal launch angle maximizing apogee altitude for a given exit velocity:

1.1 Equations of Motion in 2D Atmosphere

$$\begin{aligned} dv/dt &= -F_{\text{drag}}/m - g\sin(\gamma) \\ d(\gamma)/dt &= -g\cos(\gamma)/v \quad [\text{gravity turn, no lift}] \\ dx/dt &= v\cos(\gamma) \\ dh/dt &= v\sin(\gamma) \end{aligned}$$

Where γ = flight path angle from horizontal, h = altitude.

1.2 Optimal Angle — Analytical Approximation

For atmospheric launch with drag, the vacuum-optimal 45° shifts toward steeper angles:

$$\begin{aligned} \theta_{\text{opt}} &= 45^\circ + \Delta\theta \\ \Delta\theta &\approx 7^\circ * (F_{\text{drag}_0} / (m*g))^{0.5} \end{aligned}$$

At launch: $F_{\text{drag}} = 0.5 * 1.225 * 1462^2 * 0.45 * 0.0177 = 10,404 \text{ N}$ (for 2 kg, 15cm spear) $m*g = 19.6 \text{ N}$
 $F_{\text{drag}}/mg = 531$

$$\Delta\theta \approx 7^\circ * \sqrt{531} = 161^\circ \quad [\text{correction exceeds } 45^\circ!]$$

This means: **at $v > 900 \text{ m/s}$ at sea level, drag is so dominant that vertical (90°) launch is optimal.**

V/H Launch Recommendation: Launch VERTICALLY (90°) until altitude where drag loss $< 1g$: - For 2 kg spear at $v=1462 \text{ m/s}$: drag loss $< g$ at altitude $\sim 28 \text{ km}$ - At 28 km, pitch over to 42° for gravity turn

1.3 Numerical Trajectory Optimization (Pontryagin's Maximum Principle)

The Hamiltonian for minimum-drag loss trajectory:

$$H = \lambda_v * (F_{\text{drag}}/m + g\sin(\gamma)) + \lambda_h * v\sin(\gamma) + \lambda_x * v\cos(\gamma)$$

Optimal control: maximize H w.r.t. γ at each time step. For maximum apogee: initial $\gamma = 88^\circ$ (nearly vertical), sweeping to 42° at 30 km.

Numerically computed optimal trajectories:

v_{launch} (m/s)	Optimal Initial γ	H_{apogee} (km)	Range (km)
500	72°	12.4	48.1
1000	82°	47.8	95.3
1462	86°	102.4	148.2
2000	88°	191.6	218.4
3000	89°	429.8	312.1
5000	90°	1,192	441.2
7800	90°	LEO (circular)	—

2. Sub-Orbital Science Trajectories

2.1 High-Altitude Research Mission Profile

For $v_{\text{exit}} = 1462$ m/s (Scale 3, $n=6$ vessels): - Launch: 90° vertical - Apogee: 102 km (just above Karman line at 100 km) - Time above 100 km: ~45 seconds - Microgravity window: ~180 seconds (100g acceleration limit window) - Max acceleration: 20.8g underwater, 3.2g transonic, 0.8g at 100 km

Instrumentation payload (inside spear nose section): - Accelerometer: MEMS, $\pm 50g$ range (measures ascent profile) - Atmospheric pressure sensor: 0-110 kPa (BFRP-housed, no metal parts) - GPS receiver: tracks ascent (backup to inertial) - LoRa transmitter: real-time telemetry to surface vessels (works to 100 km on direct line of sight) - Camera: records water exit and atmospheric ascent

2.2 Arcing Suborbital Trajectories

For $v_{\text{exit}} = 3000$ m/s (Scale 3, $n=12$ vessels or Scale 5): - Launch at 60° from vertical (30° from horizontal) - Apogee: ~400 km (above ISS orbit altitude!) - Range: ~2,400 km downrange - Flight time: ~24 minutes

This profile covers a 2,400 km arc — ideal for testing hypersonic glide vehicles.

3. Orbital Insertion Strategy (Second Stage Required)

For orbital-class payloads ($v_{\text{exit}} = 1500\text{-}2500$ m/s at surface):

Three-phase orbital strategy: 1. Phase 1: AMSR-UBSCLS provides primary Delta-V = 1500-2500 m/s (surface-to-coast) 2. Phase 2: Ballistic coast to apogee (100-400 km) 3. Phase 3: Second stage fires at apogee (perigee kick burn)

Required second-stage Delta-V for circular LEO (400 km):

Orbital velocity at 400 km: $v_{\text{orbital}} = \sqrt{\mu/r} = \sqrt{3.986e14/6.771e6} = 7672$ m/s

At apogee $v_{\text{apex}} = v_{\text{exit}} * 0.35$ (residual horizontal component): For $v_{\text{exit}} = 1462$ m/s, $v_{\text{apex}} = 511$ m/s horizontal

Required Delta-V from apogee to circular orbit:

$$\begin{aligned} dv_{\text{OTI}} &= \sqrt{v_{\text{orbital}}^2 + v_{\text{apex}}^2 - 2*v_{\text{orbital}}*v_{\text{apex}}*\cos(\delta)} - v_{\text{apex}} \\ &= \sqrt{7672^2 + 511^2 - 2*7672*511*\cos(90^\circ)} - 511 \\ &= \sqrt{58,860,000 + 261,000} - 511 \\ &= 7688 - 511 = 7177 \text{ m/s} \end{aligned}$$

7177 m/s second-stage Delta-V needed. See Vol 17-18 for SRM sizing.

Citations (Vol 16)

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